

Exact PVM-Incompatibility-Breaking Criterion for some non-unital qubit channel

AI (ChatGPT 5.6 Sol, Codex 5.6 Sol)
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We study when a nonunital qubit channel renders all projective measurements jointly measurable. This note presents a self-contained channel formulation of the projective-steering results of Zhang, Kwek, and Chen. For every qubit channel whose output ellipsoid contains the origin, we recover an exact PVM-incompatibility-breaking criterion. For invertible Bloch matrices, the criterion reduces to a strictly concave optimization over three real parameters. We construct the corresponding generally asymmetric parent POVM and obtain explicit classical postprocessings for all noisy projective measurements. A matching support-function bound establishes optimality and proves that the parent at the compatibility boundary is unique up to classical refinement. The result includes translations in arbitrary directions and extends by continuity to singular Bloch matrices. We distinguish this PVM criterion from the additional conditions under which existing results establish the same boundary for arbitrary POVMs.

A quantum channel is PVM-incompatibility breaking if it maps every finite family of projective measurements to a jointly measurable family [1]. For unital qubit channels, the exact criterion is given by a spherical average of the Bloch matrix [2], and the same criterion also characterizes incompatibility breaking for arbitrary POVMs [3]. For nonunital channels, the translation breaks the central symmetry of the corresponding geometric problem. Determining an optimal parent then requires controlling both its normalization and its first moment.

The full-rank steering criterion, optimal hidden-state ensemble, and its uniqueness were established by Zhang, Kwek, and Chen [4] (Theorem S9 and Sec. S3 of their Supplemental Material). The present note offers a complementary, self-contained exposition in channel notation, with explicit parent POVMs and classical postprocessings. Appendix F gives the notation correspondence with their formulation.

Here we give a direct channel proof for channels whose output ellipsoid contains the origin. The proof compares the support function of a prescribed parent with that of the noisy projective measurements. An upper bound holds for every parent, while a vector stationarity condition determines a positive, normalized parent attaining it. The result does not require the translation to be aligned with a singular axis of the Bloch matrix.

Channel representation.— Let $\mathcal{N}_{D,\vec{t}}$ be a qubit channel whose Heisenberg adjoint has the Bloch representation

$$\mathcal{N}_{D,\vec{t}}^\dagger(x_0\mathbb{1} + \vec{x} \cdot \vec{\sigma}) = (x_0 + \vec{t} \cdot \vec{x})\mathbb{1} + (D\vec{x}) \cdot \vec{\sigma}, \quad (1)$$

where D is a real 3×3 matrix and $\vec{t} \in \mathbb{R}^3$. Thus the Schrödinger action on Bloch vectors is $\vec{r} \mapsto D^T \vec{r} + \vec{t}$. All channel statements below assume complete positivity and trace preservation.

For invertible D , the class considered here is specified by

$$\|D^{-T}\vec{t}\| \leq 1 \iff Q := D^T D - \vec{t}\vec{t}^T \geq 0. \quad (2)$$

This is equivalent to $\vec{0} \in \vec{t} + D^T \mathbb{B}^3$, where $\mathbb{B}^3$ is the closed unit Bloch ball. In particular, when the Schrödinger

Bloch matrix is denoted by T , Eq. (2) is precisely $\|T^{-1}\vec{t}\| \leq 1$, with $T = D^T$. The condition is invariant under input and output unitary rotations.

Let $d\mu(\hat{n})$ denote the normalized rotation-invariant measure on S^2 , so that $d\mu = d\Omega/(4\pi)$. For $\|\vec{k}\| < 1$, define

$$g^2 := 1 - \|\vec{k}\|^2, \quad (3a)$$

$$C_{\vec{k}} := (g^2 I + \vec{k}\vec{k}^T)^{1/2}, \quad (3a)$$

$$B_{\vec{k}} := D^T + \vec{t}\vec{k}^T, \quad (3b)$$

$$W_{D,\vec{t}}(\vec{k}) := \int_{S^2} \|B_{\vec{k}} C_{\vec{k}} \hat{n}\| d\mu(\hat{n}). \quad (3c)$$

Here I is the 3×3 identity, while $\mathbb{1}$ is the qubit identity. We write $W(\vec{k}) = W_{D,\vec{t}}(\vec{k})$ when the channel is fixed.

The definition extends continuously to $\|\vec{k}\| \leq 1$.

Exact PVM criterion.— The full-rank criterion takes the following channel form [4].

Theorem 1. Let $\mathcal{N}_{D,\vec{t}}$ be a qubit channel with invertible D satisfying Eq. (2). Then W is smooth and strictly concave on the open unit ball, and has a unique maximizer $\vec{k}_*$ in that ball. The channel is PVM-incompatibility breaking if and only if

$$2W_{D,\vec{t}}(\vec{k}_*) \leq 1. \quad (4)$$

Equivalently, its projective critical radius is

$$R_{\text{PVM}}(D, \vec{t}) = \frac{1}{2W_{D,\vec{t}}(\vec{k}_*)}. \quad (5)$$

The critical radius is defined in Appendix A. Theorem 1 replaces an optimization over parent measures by a strictly concave optimization over three real variables. It does not require an elementary expression for $\vec{k}_*$.

To state the optimal parent, evaluate $B = B_{\vec{k}_*}$ and g

at $\vec{k} = \vec{k}_*$, and write $W_* = W(\vec{k}_*)$. Define

$$\vec{b} := \vec{k} + g^2 B^{-1} \vec{t}, \quad (6)$$

$$q_B(\hat{n}) := (\hat{n} + \vec{k})^T (B^T B)^{-1} (\hat{n} + \vec{k}), \quad (7)$$

$$p_*(\hat{n}) := \frac{1 + \vec{b} \cdot \vec{k} + (\vec{b} + \vec{k}) \cdot \hat{n}}{|\det B| W_* q_B(\hat{n})^2}. \quad (8)$$

The parent POVM is $\{\Pi_*(\hat{n}) d\mu(\hat{n})\}_{\hat{n} \in S^2}$, where

$$\Pi_*(\hat{n}) := p_*(\hat{n}) (\mathbb{1} + \hat{n} \cdot \vec{\sigma}). \quad (9)$$

In Appendix C, we prove that

$$p_* \geq 0, \quad \int_{S^2} p_* d\mu = 1, \quad \int_{S^2} \hat{n} p_* d\mu = \vec{0}. \quad (10)$$

Thus Π_* is positive and normalized, but generally not centrally symmetric. If $R := R_{\text{PVM}}(D, \vec{t}) \geq 1$, it simulates every noisy PVM with postprocessing

$$p(\pm|\hat{x}, \hat{n}) = \frac{1}{2} \left[1 \pm \frac{1}{R} \text{sgn}((B^{-1} \hat{x}) \cdot (\hat{n} + \vec{k})) \right]. \quad (11)$$

Here $P_{\pm|\hat{x}} = (\mathbb{1} \pm \hat{x} \cdot \vec{\sigma})/2$ and $\hat{x} \in S^2$. At $R = 1$, the postprocessings are deterministic except on their measure-zero dividing circles.

Proof of Theorem 1. For a zero-mean probability measure ν on S^2 , define

$$F_\nu(c_0, \vec{c}) := \int_{S^2} |c_0 + \vec{c} \cdot \hat{n}| d\nu(\hat{n}). \quad (12)$$

The parent $(\mathbb{1} + \hat{n} \cdot \vec{\sigma}) d\nu(\hat{n})$ simulates all noisy PVMs if and only if

$$F_\nu(c_0, \vec{c}) \geq \|D^T \vec{c} + c_0 \vec{t}\| \quad \forall (c_0, \vec{c}). \quad (13)$$

This is the binary-measurement instance of the convex-inclusion approach to fixed-ensemble simulation [2, 5]. A self-contained proof is given in Appendix A.

For necessity, restrict the witnesses to $c_0 = \vec{k} \cdot \vec{c}$. A positive weighted average of Eq. (13) then gives $2W(\vec{k}) \leq 1$ for every $\|\vec{k}\| < 1$, independently of the parent. Appendix B proves this bound and the existence of a unique interior maximizer.

For sufficiency, the stationarity condition at $\vec{k}_*$ implies Eq. (10). Moreover, the corresponding measure $d\nu_* = p_* d\mu$ satisfies

$$F_{\nu_*}(c_0, \vec{c}) \geq \frac{\|D^T \vec{c} + c_0 \vec{t}\|}{2W_*} \quad \forall (c_0, \vec{c}), \quad (14)$$

with equality whenever $c_0 = \vec{k}_* \cdot \vec{c}$. Appendix C establishes this inequality by pairing antipodal directions and applying convexity of the absolute value. It also verifies Eq. (11) directly. Therefore $2W_* \leq 1$ is sufficient, and the matching bounds give Eq. (5). $\square$

Uniqueness of the boundary parent.– The equality conditions in the preceding argument also constrain every optimal parent [4].

Theorem 2. Under the assumptions of Theorem 1, ν_* is the unique zero-mean probability measure on S^2 attaining the projective critical radius. If $2W_* = 1$, every parent simulating all noisy PVMs has rank-one operator density almost everywhere and induces ν_* after aggregation by Bloch direction. On standard Borel outcome spaces, the parent is therefore unique up to classical refinement and relabelling. For each PVM direction, the postprocessing is unique almost everywhere.

The proof is given in Appendix D. The result includes $\|D^{-T} \vec{t}\| = 1$. It is a boundary statement about all simulating parents; it does not assert uniqueness of arbitrary parents in the interior of the PVM-incompatibility-breaking region.

Singular channels and special cases.– The inverse-free condition $Q \geq 0$ also allows singular Bloch matrices.

Corollary 1. Let $\mathcal{N}_{D, \vec{t}}$ be a qubit channel satisfying $D^T D - \vec{t} \vec{t}^T \geq 0$, without assuming that D is invertible. Then it is PVM-incompatibility breaking if and only if

$$2 \max_{\|\vec{k}\| \leq 1} W_{D, \vec{t}}(\vec{k}) \leq 1. \quad (15)$$

For $(D, \vec{t}) \neq (0, \vec{0})$, the reciprocal of the left-hand side is the projective critical radius. An optimal parent is obtained as a weak limit of the nonsingular optimal measures.

Appendix E gives a continuity proof. The limiting measure may be singular; neither the density in Eq. (8) nor the uniqueness conclusion is asserted for singular D .

For $\vec{t} = \vec{0}$, symmetry and strict concavity give $\vec{k}_* = \vec{0}$, and Theorem 1 reduces to

$$2m_D \leq 1, \quad m_D := \int_{S^2} \|D\hat{n}\| d\mu(\hat{n}). \quad (16)$$

This is the channel form of the T -state criterion obtained from the ellipsoidal local-hidden-state model [6] and its optimality proof [7]. Under the change of outcome $\hat{u} = D\hat{n}/\|D\hat{n}\|$, the optimal rank-one parent becomes

$$\tilde{\Pi}_*(\hat{n}) = \frac{1}{m_D} [\|D\hat{n}\| \mathbb{1} + (D\hat{n}) \cdot \vec{\sigma}]. \quad (17)$$

At $m_D = 1/2$, this is precisely the boundary parent $2[\|D\hat{n}\| \mathbb{1} + (D\hat{n}) \cdot \vec{\sigma}]$ used in Ref. [3]. Away from the boundary, different parent representations and postprocessings need not coincide.

For $D = \text{diag}(d_\perp, d_\perp, d_z)$ and $\vec{t} = \tau \hat{e}_z$, with $d_\perp, d_z > 0$, the hypothesis becomes $|\tau| \leq d_z$. Rotational symmetry gives $\vec{k}_* = k_* \hat{e}_z$, reducing the optimization to

$$W(k) = \int_0^1 \sqrt{d_\perp^2 (1 - k^2)(1 - z^2) + (d_z + \tau k)^2 z^2} dz. \quad (18)$$

For general invertible diagonal $D = \text{diag}(d_1, d_2, d_3)$, the condition is $\sum_i t_i^2/d_i^2 \leq 1$, and all three components of $\vec{t}$ may be nonzero.

The spherical average also admits an elliptic-integral representation,

$$W(\vec{k}) = R_G(\ell_1, \ell_2, \ell_3), \quad (19)$$

where ℓ_i are the eigenvalues of $B_{\vec{k}} C_k^2 B_{\vec{k}}^T$ and R_G is Carlson's symmetric elliptic integral of the second kind. Equation (19) follows from its spherical integral representation [8]. Thus the exact boundary is determined by $2W(\vec{k}_*) = 1$ together with $\nabla W(\vec{k}_*) = \vec{0}$. Along a ray $(sD, s\vec{t})$ with $s > 0$, the critical radius scales as $1/s$ and the optimal parent is unchanged; only the CPTP portion of the ray describes physical channels.

Conclusion. – We have presented a channel formulation

of the projective-steering criterion established by Zhang, Kwek, and Chen [4], for qubit channels whose output ellipsoid contains the origin. For invertible Bloch matrices, a vector stationarity identity yields an explicit optimal parent, while a matching support-function bound establishes necessity and uniqueness at the boundary. A continuity argument extends the criterion to singular matrices.

The remaining questions are distinct. If $\|D^{-T}\vec{t}\| > 1$, the parent construction need not be positive, and the present proof does not determine the exact PVM boundary. The PVM result alone does not establish incompatibility breaking for arbitrary POVMs. Beyond the unital result [3], Ref. [4] proves equality of the PVM and POVM scales for all correlation-rank-at-most-two data (Theorem S11), and for explicit full-rank bias domains (Theorem S12). Equality throughout the full-rank class in Eq. (2) is not established here.

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Appendix A: Support-function characterization and critical radius

We first justify Eq. (13), including the existence of a common parent for all projective directions. For a prob-

ability measure ν on S^2 with $\int \hat{n} d\nu(\hat{n}) = \vec{0}$, consider the convex set

$$Z_\nu := \left\{ \int_{S^2} f(\hat{n})(1, \hat{n}) d\nu(\hat{n}) : f \in L^\infty(\nu), |f| \leq 1 \right\}. \quad (A1)$$

The unit ball of $L^\infty(\nu)$ is weak-star compact, and the four moment maps are weak-star continuous. Thus Z_ν is compact. Its support function is F_ν in Eq. (12).

A binary postprocessing of the parent $(\mathbb{1} + \hat{n} \cdot \vec{\sigma})d\nu(\hat{n})$ corresponds to $f = p(+|\hat{x}, \hat{n}) - p(-|\hat{x}, \hat{n}) \in [-1, 1]$. It reproduces $\mathcal{N}_{D, \vec{t}}^\dagger(P_{\pm|\hat{x}})$ precisely when

$$\int f(\hat{n})(1, \hat{n}) d\nu(\hat{n}) = (\vec{t} \cdot \hat{x}, D\hat{x}). \quad (A2)$$

Consequently, simulation for all PVMs is equivalent to the inclusion

$$\{(\vec{t} \cdot \vec{x}, D\vec{x}) : \|\vec{x}\| \leq 1\} \subseteq Z_\nu. \quad (A3)$$

The support function of the set on the left is $\|D^T \vec{c} + c_0 \vec{t}\|$, proving Eq. (13).

There is no loss in restricting the existence problem to rank-one parents [5]. Any Bloch vector in the closed ball admits a probability distribution on the sphere with the same mean. Applying this refinement to each parent effect preserves every postprocessing. For each finite PVM family, the set of zero-mean sphere probabilities simulating that family is closed, because it is specified by continuous support inequalities. These closed subsets of a compact space have the finite intersection property whenever every finite PVM family is compatible. Their intersection is therefore nonempty, giving a common parent for all directions.

For arbitrary real data $(D, \vec{t}) \neq (0, \vec{0})$, define

$$r_\nu(D, \vec{t}) := \inf_{\|D^T \vec{c} + c_0 \vec{t}\| > 0} \frac{F_\nu(c_0, \vec{c})}{\|D^T \vec{c} + c_0 \vec{t}\|}, \quad (\text{A4a})$$

$$R_{\text{PVM}}(D, \vec{t}) := \sup_{\substack{\nu \text{ a probability on } S^2 \\ \int \hat{n} d\nu = \vec{0}}} r_\nu(D, \vec{t}). \quad (\text{A4b})$$

The unrestricted witness convention in Eq. (A4a) will be used throughout. The relevant support inequalities are closed under weak convergence, so the supremum is attained. For CPTP data, the channel is PVM-incompatibility breaking exactly when $R_{\text{PVM}} \geq 1$. This is the channel formulation of the critical-radius geometry [2, 7]. The identification of parent models with local-hidden-state models uses the steering-joint-measurability correspondence [9].

Appendix B: Variational problem and parent-independent upper bound

1. Spherical integral identities

The density below is the ellipsoidal ansatz used for T -state local-hidden-state models [6, 7]. We give a direct change-of-variables derivation valid also for a nonsymmetric matrix. For an invertible real matrix B , define

$$m_B := \int_{S^2} \|B\hat{x}\| d\mu(\hat{x}),$$

$$w_B(\hat{u}) := \|B^{-T}\hat{u}\|^{-4}, \quad \rho_B(\hat{u}) := \frac{w_B(\hat{u})}{|\det B| m_B}. \quad (\text{B1})$$

The change of variables

$$\hat{u} = \frac{B^T \hat{x}}{\|B^T \hat{x}\|}, \quad d\mu(\hat{u}) = \frac{|\det B|}{\|B^T \hat{x}\|^3} d\mu(\hat{x}) \quad (\text{B2})$$

gives

$$\int_{S^2} \rho_B d\mu = 1, \quad (\text{B3a})$$

$$\int_{S^2} \rho_B(\hat{u}) |\vec{c} \cdot \hat{u}| d\mu(\hat{u}) = \frac{\|B\vec{c}\|}{2m_B}, \quad (\text{B3b})$$

$$\int_{S^2} \rho_B(\hat{u}) \|C\hat{u}\| d\mu(\hat{u}) = \frac{1}{m_B} \int_{S^2} \|CB^T \hat{x}\| d\mu(\hat{x}). \quad (\text{B3c})$$

Indeed, writing $\ell = \|B^T \hat{x}\|$ gives $w_B(\hat{u}) = \ell^4$. The integrand in Eq. (B3b), including its Jacobian, then reduces to $|(B\vec{c}) \cdot \hat{x}|/(m_B)$, whose spherical average is $\|B\vec{c}\|/(2m_B)$. Moreover, $m_B = m_{B^T}$ since the average depends only on the singular values. For symmetric $C = C_k$, one has $CB^T = (BC)^T$, so

$$\int_{S^2} \rho_B(\hat{u}) \|C_k \hat{u}\| d\mu(\hat{u}) = \frac{W(\vec{k})}{m_B}. \quad (\text{B4})$$

No commutativity of B and C_k is required.

2. Strict concavity and the interior maximizer

Let $\vec{\xi} = D^{-T} \vec{t}$, with $\|\vec{\xi}\| \leq 1$. The matrix determinant lemma gives

$$\det B_{\vec{k}} = \det D (1 + \vec{k} \cdot \vec{\xi}) \neq 0 \quad (\|\vec{k}\| < 1). \quad (\text{B5})$$

Direct multiplication gives

$$B_{\vec{k}} C_k^2 B_{\vec{k}}^T = g^2 Q + (D^T \vec{k} + \vec{t})(D^T \vec{k} + \vec{t})^T. \quad (\text{B6})$$

Consequently,

$$W(\vec{k}) = \int_{S^2} \sqrt{g^2 \hat{x}^T Q \hat{x} + [\hat{x} \cdot (D^T \vec{k} + \vec{t})]^2} d\mu(\hat{x}). \quad (\text{B7})$$

For fixed $\hat{x} \in S^2$, let $A = \|D\hat{x}\| > 0$, $h = \vec{t} \cdot \hat{x}$, and $\hat{e} = D\hat{x}/A$. Writing $\vec{k} = \zeta \hat{e} + \vec{k}_\perp$, the integrand in Eq. (B7) becomes

$$\sqrt{(A + h\zeta)^2 - (A^2 - h^2) \|\vec{k}_\perp\|^2}, \quad (\text{B8})$$

where $A^2 - h^2 = \hat{x}^T Q \hat{x} \geq 0$. The function $(L, \vec{z}) \mapsto \sqrt{L^2 - \|\vec{z}\|^2}$ is concave on $L > \|\vec{z}\|$. For a variation $(a, \vec{w})$, its second derivative is

$$-\frac{\|L\vec{w} - a\vec{z}\|^2 - (\|\vec{z}\|^2 \|\vec{w}\|^2 - (\vec{z} \cdot \vec{w})^2)}{(L^2 - \|\vec{z}\|^2)^{3/2}} \leq 0. \quad (\text{B9})$$

To see the sign, decompose $\vec{w}$ parallel and perpendicular to $\vec{z}$. The numerator is a square plus $(L^2 - \|\vec{z}\|^2) \|\vec{w}_\perp\|^2$. The arguments in Eq. (B8) lie in this domain: $A + h\zeta > 0$ and the radicand is positive by the invertibility of $B_{\vec{k}}$ and C_k . Thus W is concave.

For strict concavity, fix any nonzero direction $\vec{v}$. Since $Q = D^T(I - \vec{\xi}\vec{\xi}^T)D$ has rank at least two, there exists $\hat{x}$ with $\hat{x} \cdot D^T \vec{v} = 0$ and $\hat{x}^T Q \hat{x} > 0$. The variation of $A + h\zeta$ in direction $\vec{v}$ then vanishes, whereas the perpendicular variation is nonzero. The second derivative in Eq. (B8) is therefore strictly negative. It remains negative on a neighborhood of positive spherical measure, implying

$$\vec{v}^T \nabla^2 W(\vec{k}) \vec{v} < 0 \quad (\vec{v} \neq \vec{0}). \quad (\text{B10})$$

Smoothness follows from a positive lower bound for the radicand on each compact subset of the open ball.

It remains to exclude a maximum on the boundary. For $\hat{e} \in S^2$, put $\vec{h} = D^T \hat{e} + \vec{t}$. Equation (B7) gives $W(\hat{e}) = \|\vec{h}\|/2$. If $\vec{h} = \vec{0}$, this is smaller than $W(\vec{0})$. Otherwise, set $\vec{k} = (1 - \epsilon)\hat{e}$ and $\delta = 2\epsilon - \epsilon^2$. The Lipschitz property of the Euclidean norm gives

$$\begin{aligned} & W((1 - \epsilon)\hat{e}) - W(\hat{e}) \\ & \geq \int_{S^2} \left[\sqrt{(\hat{x} \cdot \vec{h})^2 + \delta \hat{x}^T Q \hat{x} - |\hat{x} \cdot \vec{h}|} \right] d\mu(\hat{x}) \\ & \quad - \epsilon \|D^T \hat{e}\|. \end{aligned} \quad (\text{B11})$$

Because rank $Q \geq 2$, an arc of the equator $\hat{x} \cdot \vec{h} = 0$ has a neighborhood on which $\hat{x}^T Q \hat{x} \geq q_0 > 0$. In coordinates $z = \hat{x} \cdot \vec{h} / \|\vec{h}\|$ and azimuth, the bracketed integrand is bounded below by $c\delta/|z|$ on $\sqrt{\delta} \leq |z| \leq z_0$, for fixed positive c, z_0 . This follows by multiplying the difference of square roots by its conjugate. The integral is therefore at least $c'\delta \log(1/\delta) - O(\delta)$, which dominates the last term of Eq. (B11). Every boundary point thus admits an inward improvement. Compactness and strict concavity establish the unique interior maximizer, including when $\|\vec{\xi}\| = 1$.

3. Upper bound for arbitrary parents

Fix $\|\vec{k}\| < 1$ and a zero-mean sphere probability ν . Restrict $F_\nu(c_0, \vec{c}) \geq r_\nu \|D^T \vec{c} + c_0 \vec{t}\|$ to $c_0 = \vec{k} \cdot \vec{c}$, and average over $\vec{c} = \hat{c} \in S^2$ with the positive weight

$$\omega_C(\hat{c}) := \|C^{-1} \hat{c}\|^{-4}, \quad C = C_{\vec{k}}. \quad (\text{B12})$$

Equation (B3b) gives

$$\begin{aligned} \int_{S^2} \omega_C(\hat{c}) F_\nu(\vec{k} \cdot \hat{c}, \hat{c}) d\mu(\hat{c}) \\ = \frac{\det C}{2} \int_{S^2} \|C(\hat{n} + \vec{k})\| d\nu(\hat{n}) = \frac{\det C}{2}, \end{aligned} \quad (\text{B13})$$

where the last equality follows from

$$\|C(\hat{n} + \vec{k})\| = 1 + \vec{k} \cdot \hat{n} \quad (\hat{n} \in S^2) \quad (\text{B14})$$

and $\int \hat{n} d\nu = \vec{0}$. The same change of variables gives

$$\int_{S^2} \omega_C(\hat{c}) \|B_{\vec{k}} \hat{c}\| d\mu(\hat{c}) = \det C W(\vec{k}). \quad (\text{B15})$$

Hence

$$r_\nu(D, \vec{t}) \leq \frac{1}{2W(\vec{k})}, \quad R_{\text{PVM}}(D, \vec{t}) \leq \frac{1}{2W_*}. \quad (\text{B16})$$

The weighted-average argument does not require $Q \geq 0$. That assumption enters the construction of the matching positive parent.

Appendix C: Construction of the optimal parent

1. Vector stationarity condition

For an arbitrary interior $\vec{k}$, write $B = B_{\vec{k}}$ and define

$$s(\hat{u}) := \sqrt{g^2 + (\vec{k} \cdot \hat{u})^2} = \|C\hat{u}\|, \quad (\text{C1a})$$

$$I_s := \int_{S^2} \rho_B(\hat{u}) s(\hat{u}) d\mu(\hat{u}) = \frac{W}{m_B}, \quad (\text{C1b})$$

$$J := \int_{S^2} \rho_B(\hat{u}) \frac{\hat{u} \hat{u}^T}{s(\hat{u})} d\mu(\hat{u}), \quad (\text{C1c})$$

$$\vec{v} := B^{-1} \vec{t}, \quad \vec{b} := \vec{k} + g^2 \vec{v}. \quad (\text{C1d})$$

We claim that

$$\nabla_{\vec{k}} W = \frac{m_B}{g^2} C^2 (J\vec{b} - I_s \vec{k}). \quad (\text{C2})$$

To prove this, write $\vec{y} = B^T \hat{x} = \ell \hat{u}$ in $W = \int \sqrt{\vec{y}^T C^2 \vec{y}} d\mu(\hat{x})$. Since $d\vec{y} = (\vec{t} \cdot \hat{x}) d\vec{k} = \ell(\vec{v} \cdot \hat{u}) d\vec{k}$, differentiation gives

$$\nabla W = \int_{S^2} \frac{\ell}{s} [(\vec{b} \cdot \hat{u}) \hat{u} - \vec{k} \{1 - (\vec{v} \cdot \hat{u})(\vec{k} \cdot \hat{u})\}] d\mu(\hat{x}). \quad (\text{C3})$$

Equation (B2) implies $\ell d\mu(\hat{x}) = m_B \rho_B(\hat{u}) d\mu(\hat{u})$. Let $K_0 = \int \rho_B/s d\mu$; then

$$\begin{aligned} I_s &= g^2 K_0 + \vec{k}^T J \vec{k}, \\ K_0 - \vec{k}^T J \vec{v} &= \frac{I_s - \vec{k}^T J \vec{b}}{g^2}. \end{aligned} \quad (\text{C4})$$

Substituting these identities into Eq. (C3), and using $C^2 = g^2 I + \vec{k} \vec{k}^T$ and $C^2 \vec{k} = \vec{k}$, proves Eq. (C2). Thus stationarity at $\vec{k}_*$ is equivalent to

$$J \vec{b} = I_s \vec{k}. \quad (\text{C5})$$

This vector equation supplies all three first-moment constraints; no coordinate-reflection symmetry is needed.

2. Positivity and normalization

From now on set $\vec{k} = \vec{k}_*$. Define

$$r(\hat{u}) := s(\hat{u}) + \vec{k} \cdot \hat{u}, \quad \hat{n}(\hat{u}) := r(\hat{u}) \hat{u} - \vec{k}. \quad (\text{C6})$$

Since $r > 0$ and $\|\hat{n}\| = 1$, this is a smooth bijection of S^2 , with inverse $\hat{u} = (\hat{n} + \vec{k}) / \|\hat{n} + \vec{k}\|$. Define a measure η in the $\hat{u}$ coordinates by

$$d\eta(\hat{u}) := \frac{\rho_B(\hat{u})}{I_s r(\hat{u})} \left(1 + \frac{\vec{b} \cdot \hat{u}}{s(\hat{u})}\right) d\mu(\hat{u}), \quad (\text{C7})$$

and let ν_* be its pushforward under Eq. (C6).

For $\vec{\xi} = D^{-T}\vec{t}$, the Sherman–Morrison formula gives

$$\vec{v} = \frac{\vec{\xi}}{1 + \vec{k} \cdot \vec{\xi}}, \quad C^{-2} = \frac{I - \vec{k}\vec{k}^T}{g^2}. \quad (\text{C8})$$

Substituting $\vec{b} = \vec{k} + g^2\vec{v}$ yields

$$1 - \vec{b}^T C^{-2} \vec{b} = \frac{g^2(1 - \|\vec{\xi}\|^2)}{(1 + \vec{k} \cdot \vec{\xi})^2} \geq 0. \quad (\text{C9})$$

Therefore,

$$|\vec{b} \cdot \hat{u}| \leq \|C^{-1}\vec{b}\| \|C\hat{u}\| \leq s(\hat{u}), \quad (\text{C10})$$

proving positivity of η and ν_* . If $\|\vec{\xi}\| < 1$, the density is strictly positive. If $\|\vec{\xi}\| = 1$, equality in Cauchy–Schwarz fixes $\hat{u}$ up to sign, and the density vanishes only at the point with the negative sign. Thus the endpoint is also included.

Put $a(\hat{u}) = \vec{k} \cdot \hat{u}$. Since $s^2 - a^2 = g^2$,

$$\frac{1}{r} = \frac{s - a}{g^2}. \quad (\text{C11})$$

Evenness of ρ_B and s then gives

$$\int_{S^2} d\nu_* = \frac{I_s - \vec{k}^T J \vec{b}}{I_s g^2} = 1, \quad (\text{C12})$$

$$\int_{S^2} \hat{n} d\nu_* = \frac{J \vec{b}}{I_s} - \vec{k} = \vec{0}, \quad (\text{C13})$$

where the last equalities use Eq. (C5).

To obtain the density on the original Bloch sphere, use

$$\begin{aligned} d\mu(\hat{u}) &= \frac{s}{r^2} d\mu(\hat{n}), \\ rs &= 1 + \vec{k} \cdot \hat{n}. \end{aligned} \quad (\text{C14})$$

Also $w_B(\hat{u}) = r^4/q_B(\hat{n})^2$. Inserting these identities into Eq. (C7), and using $m_B I_s = W_*$, gives $d\nu_*(\hat{n}) = p_*(\hat{n})d\mu(\hat{n})$ with p_* in Eq. (8). In particular, the quadratic form involves $(B^T B)^{-1}$, consistently with the definition $B = D^T + \vec{t}\vec{k}^T$.

3. Matching the support inequalities

For any witness $(c_0, \vec{c})$, put $d = c_0 - \vec{k} \cdot \vec{c}$ and $y = \vec{c} \cdot \hat{u}$. Equation (C7) gives

$$F_{\nu_*}(c_0, \vec{c}) = \frac{1}{I_s} \int_{S^2} \rho_B \left(1 + \frac{\vec{b} \cdot \hat{u}}{s} \right) \left| y + \frac{d(s-a)}{g^2} \right| d\mu(\hat{u}). \quad (\text{C15})$$

Pair $\hat{u}$ and $-\hat{u}$. The weights $(1 \pm \vec{b} \cdot \hat{u}/s)/2$ are nonnegative, so convexity of the absolute value gives

$$\begin{aligned} & \frac{1 + \vec{b} \cdot \hat{u}/s}{2} \left| y + \frac{d(s-a)}{g^2} \right| + \frac{1 - \vec{b} \cdot \hat{u}/s}{2} \left| y - \frac{d(s-a)}{g^2} \right| \\ & \geq \left| y + \frac{d(\vec{b} - \vec{k}) \cdot \hat{u}}{g^2} \right| = |(\vec{c} + dB^{-1}\vec{t}) \cdot \hat{u}|. \end{aligned} \quad (\text{C16})$$

The cosine identity in Eq. (B3b) therefore yields

$$\begin{aligned} F_{\nu_*}(c_0, \vec{c}) & \geq \frac{\|B\vec{c} + d\vec{t}\|}{2m_B I_s} \\ & = \frac{\|D^T \vec{c} + c_0 \vec{t}\|}{2W_*}. \end{aligned} \quad (\text{C17})$$

Equality holds whenever $d = 0$, that is, on the entire hyperplane

$$c_0 = \vec{k}_* \cdot \vec{c}. \quad (\text{C18})$$

Thus $r_{\nu_*} = 1/(2W_*)$, matching Eq. (B16).

4. Explicit classical postprocessings

For $\hat{x} \in S^2$, let $\vec{c} = B^{-1}\hat{x}$ and

$$f_{\hat{x}}(\hat{n}) := \text{sgn}[\vec{c} \cdot (\hat{n} + \vec{k})]. \quad (\text{C19})$$

Differentiating Eq. (B3b) with respect to $\vec{c}$ gives

$$\int_{S^2} \rho_B(\hat{u}) \hat{u} \text{sgn}(\vec{c} \cdot \hat{u}) d\mu(\hat{u}) = \frac{B^T B \vec{c}}{2m_B \|B\vec{c}\|} = \frac{B^T \hat{x}}{2m_B}. \quad (\text{C20})$$

The derivative exists because ρ_B is smooth and great circles have zero measure. Let $d\zeta = r d\eta$. The even part of ζ and the odd part of η satisfy

$$\begin{aligned} d\zeta_{\text{even}} &= \frac{\rho_B}{I_s} d\mu, \\ d\eta_{\text{odd}} &= \frac{\rho_B}{I_s} (\vec{v} \cdot \hat{u}) d\mu. \end{aligned} \quad (\text{C21})$$

The second identity follows from Eq. (C11) and $(\vec{b} - \vec{k})/g^2 = \vec{v}$. With $R = 1/(2W_*)$, this implies

$$\int_{S^2} f_{\hat{x}} d\nu_* = R \vec{t} \cdot \hat{x}, \quad (\text{C22})$$

$$\int_{S^2} \hat{n} f_{\hat{x}} d\nu_* = R B^T \hat{x} - \vec{k} R (\vec{t} \cdot \hat{x}) = R D \hat{x}. \quad (\text{C23})$$

Consequently $p(\pm|\hat{x}, \hat{n}) = (1 \pm f_{\hat{x}}/R)/2$ reproduces $\mathcal{N}_{D, \vec{t}}^\dagger(P_{\pm|\hat{x}})$ and lies in $[0, 1]$ whenever $R \geq 1$, proving Eq. (11).

Appendix D: Proof of Theorem 2

1. Uniqueness of the optimal sphere measure

Proof. Let ν_1, ν_2 be optimal zero-mean sphere probabilities and write $R = 1/(2W_*)$. Their nonnegative continuous slack in the proof of Eq. (B16) has zero integral against the strictly positive weight ω_C . Hence

$$F_{\nu_i}(\vec{k} \cdot \vec{c}, \vec{c}) = R \|B\vec{c}\| \quad \forall \vec{c}, \quad (\text{D1})$$

where $\vec{k} = \vec{k}_*$. Pull the two measures back to the $\hat{u}$ sphere under Eq. (C6), obtaining η_i , and define $d\zeta_i = r d\eta_i$. Equation (D1) fixes the spherical cosine transform of ζ_i . Injectivity on even finite signed measures implies

$$d(\zeta_i)_{\text{even}} = \frac{\rho_B}{I_s} d\mu. \quad (\text{D2})$$

The injectivity statements used here are verified below.

Since ζ_i are positive and their even parts are absolutely continuous, each ζ_i is absolutely continuous. Indeed, a μ -null set and its antipode have zero mass for the even part, and therefore for the original positive measure. The same holds for η_i , because r is bounded away from zero. Thus differentiation in c_0 at the active hyperplane is legitimate. Tangency of $F_{\nu_i}(c_0, \vec{c}) - R \|D^T \vec{c} + c_0 \vec{t}\| \geq 0$ at its zero gives

$$\int_{S^2} \text{sgn}(\vec{c} \cdot \hat{u}) d\eta_i(\hat{u}) = R \frac{\vec{t} \cdot B\vec{c}}{\|B\vec{c}\|} \quad (\vec{c} \neq \vec{0}). \quad (\text{D3})$$

The sign transform is injective on odd finite signed measures, so $\eta_1 - \eta_2$ is even. On the other hand, $\zeta_1 - \zeta_2$ is odd, and

$$d(\zeta_1 - \zeta_2) = (s + \vec{k} \cdot \hat{u}) d(\eta_1 - \eta_2). \quad (\text{D4})$$

Its even part is $s d(\eta_1 - \eta_2)$, which must vanish. Since $s \geq g > 0$, the two measures coincide. This proves uniqueness of the sphere measure. $\square$

For completeness, the Funk–Hecke formula [10] diagonalizes the cosine and sign transforms on spherical harmonics. Up to a common normalization factor, their relevant eigenvalues are

$$\int_{-1}^1 |z| P_{2j}(z) dz = -\frac{2P_{2j}(0)}{(2j-1)(2j+2)}, \quad (\text{D5})$$

$$\int_{-1}^1 \text{sgn}(z) P_{2j+1}(z) dz = \frac{P_{2j}(0)}{j+1}, \quad (\text{D6})$$

$$P_{2j}(0) = (-1)^j \binom{2j}{j} 4^{-j} \neq 0. \quad (\text{D7})$$

These identities follow from the Legendre recurrence and integration by parts. All required eigenvalues are nonzero. Vanishing transform therefore annihilates every spherical harmonic of the appropriate parity. Density of spherical polynomials in continuous functions proves injectivity for finite signed measures.

2. General parents and boundary postprocessings

Proof. For a general qubit parent on an outcome space Λ , take its normalized trace measure λ . Its Radon–Nikodym density has the form

$$\Pi(d\omega) = [\mathbb{1} + \vec{v}(\omega) \cdot \vec{\sigma}] d\lambda(\omega), \quad \|\vec{v}(\omega)\| \leq 1. \quad (\text{D8})$$

Push λ forward to the closed Bloch ball. The numerator in Eq. (B13) now contains $\|C(\vec{v} + \vec{k})\|$, and

$$(1 + \vec{k} \cdot \vec{v})^2 - \|C(\vec{v} + \vec{k})\|^2 = g^2(1 - \|\vec{v}\|^2). \quad (\text{D9})$$

The right-hand side is strictly positive for interior Bloch vectors. At $R = 1$, simulation saturates the upper bound, forcing $\|\vec{v}\| = 1$ almost everywhere. Aggregation by Bloch direction therefore yields the unique sphere measure already proved. On standard Borel outcome spaces, any additional labels are classical conditional refinements, so these presentations are postprocessing-equivalent.

For a fixed PVM direction $\hat{x}$, a contact witness is

$$(c_0, \vec{c}) = (\vec{k} \cdot B^{-1}\hat{x}, B^{-1}\hat{x}). \quad (\text{D10})$$

If $f \in [-1, 1]$ reproduces that measurement at $R = 1$, equality in the support inequality requires

$$f(\hat{n}) = \text{sgn}[(B^{-1}\hat{x}) \cdot (\hat{n} + \vec{k})] \quad (\text{D11})$$

except on the dividing circle. This circle has zero measure because ν_* is absolutely continuous. The postprocessing is therefore unique almost everywhere. $\square$

Appendix E: Proof of Corollary 1

Proof. For singular D , the inverse-free hypothesis is equivalently

$$\vec{t} \in \text{Ran}(D^T), \quad \|(D^T)^+ \vec{t}\| \leq 1, \quad (\text{E1})$$

where $+$ denotes the Moore–Penrose inverse. We prove continuity through a homogeneous parent cost, rather than presuming continuity of the max–min expression in Eq. (A4b). For arbitrary real Bloch data, define

$$\mathcal{C}(D, \vec{t}) := \inf \{ \lambda(S^2) : \lambda \geq 0, \int \hat{n} d\lambda = \vec{0}, F_\lambda(c_0, \vec{c}) \geq \|D^T \vec{c} + c_0 \vec{t}\| \quad \forall (c_0, \vec{c}) \}. \quad (\text{E2})$$

The feasible set is nonempty, and the infimum is attained by weak compactness on bounded-mass measures. For nonzero data, normalization gives $\mathcal{C} = 1/R_{\text{PVM}}$.

For the normalized uniform measure μ ,

$$F_\mu(c_0, \vec{c}) \geq |c_0|, \quad F_\mu(c_0, \vec{c}) \geq \frac{\|\vec{c}\|}{2}. \quad (\text{E3})$$

The first bound follows from Jensen's inequality. The second follows because F_μ is even and convex in c_0 , with minimum $\|\vec{c}\|/2$ at zero. Adding $[2\|\Delta D\|_{\text{op}} + \|\Delta\vec{t}\|]\mu$ to a feasible measure and using the triangle inequality gives

$$|\mathcal{C}(D + \Delta D, \vec{t} + \Delta\vec{t}) - \mathcal{C}(D, \vec{t})| \leq 2\|\Delta D\|_{\text{op}} + \|\Delta\vec{t}\|. \quad (\text{E4})$$

Thus the cost is Lipschitz continuous in all real Bloch data.

Choose a real singular-value decomposition $D^T = U \text{diag}(s_i) V^T$, set $\vec{\xi} = (D^T)^+ \vec{t}$, and define

$$\begin{aligned} D_\epsilon^T &:= U \text{diag}(s_i + \epsilon) V^T, \\ \vec{t}_\epsilon &:= D_\epsilon^T \vec{\xi}. \end{aligned} \quad (\text{E5})$$

Then D_ϵ is invertible, $\|D_\epsilon^{-T} \vec{t}_\epsilon\| = \|\vec{\xi}\| \leq 1$, and the data converge to $(D, \vec{t})$. The nonsingular geometric proof does not use complete positivity, so it applies to these approximants even if they are not CPTP. It gives

$$\mathcal{C}(D_\epsilon, \vec{t}_\epsilon) = 2 \max_{\|\vec{k}\| \leq 1} W_{D_\epsilon, \vec{t}_\epsilon}(\vec{k}). \quad (\text{E6})$$

The functions on the right converge uniformly on the compact closed ball, and therefore their maxima converge. Equation (E4) yields

$$\mathcal{C}(D, \vec{t}) = 2 \max_{\|\vec{k}\| \leq 1} W_{D, \vec{t}}(\vec{k}), \quad (\text{E7})$$

proving the criterion and the critical-radius formula. A weakly convergent subsequence of the normalized optimal measures attains the limiting radius, by passing to the limit in every support inequality. Finally, if $D = 0$ and $Q \geq 0$, then $\vec{t} = 0$: the channel is completely depolarizing, with $\mathcal{C} = 0$ and $R_{\text{PVM}} = \infty$. $\square$

Appendix F: Correspondence with the steering formulation

Use the normalized Choi state

$$\begin{aligned} \rho_{\mathcal{N}} &= (\mathcal{N}_{D, \vec{t}} \otimes \text{id})(|\Phi^+\rangle\langle\Phi^+|), \\ |\Phi^+\rangle &= (|00\rangle + |11\rangle)/\sqrt{2}. \end{aligned} \quad (\text{F1})$$

Its second marginal is $\mathbb{1}/2$. Put $S_0 = \text{diag}(1, -1, 1)$, so that transposition sends $(\vec{n} \cdot \vec{\sigma})^T = (S_0 \vec{n}) \cdot \vec{\sigma}$. The Bloch data and contact coordinate of Ref. [4] are

$$\vec{a} = \vec{t}, \quad C = D^T S_0, \quad \vec{q} = S_0 \vec{k}. \quad (\text{F2})$$

Consequently, direct substitution gives

$$\begin{aligned} J_{\text{ref}} &= C C^T - \vec{a} \vec{a}^T = Q, \\ M_{\text{ref}}(S_0 \vec{k}) &= B_{\vec{k}} C_k^2 B_{\vec{k}}^T, \\ \Phi_{\text{ref}}(S_0 \vec{k}) &= 2W_{D, \vec{t}}(\vec{k}), \\ c_{\text{P}} &= 2W_* = R_{\text{PVM}}(D, \vec{t})^{-1}. \end{aligned} \quad (\text{F3})$$

Equations (8)–(9) and Theorem S9 of Ref. [4] therefore yield Eqs. (4)–(5) in channel notation, including the semidefinite boundary of $Q \geq 0$.

Let μ_*^{ref} be their optimal spectrum, of mass c_{P} . Reflection and normalization give

$$\nu_* = (S_0)_\# \mu_*^{\text{ref}} / c_{\text{P}}, \quad (\text{F4})$$

where $(S_0)_\#$ denotes pushforward. This follows from the support inequalities and uniqueness. Moreover,

$$C + \vec{a} \vec{a}^T = B_{\vec{k}_*} S_0. \quad (\text{F5})$$

The projective response constructed in Sec. S3 of Ref. [4] thus agrees with Eq. (11) at the boundary. To distinguish the conventions, the Lorentz-transformed matrix B in that reference differs from $B_{\vec{k}}$ defined in Eq. (3b). The appendices provide a direct verification using antipodal pairing and explain the equivalence of general boundary parents under classical refinement.